Physics-Informed Digital Twin and Advanced Ensemble Learning for Electric Vehicle Battery Prognostics and Driving Range Optimization

Course: 19MAM56 – Advanced Machine Learning Lab (CAT II Project Work)
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Abstract— *Electric Vehicle (EV) adoption relies heavily on predictable driving range and reliable battery lifespan forecasting. However, real-world battery degradation exhibits non-linear kinetics under varying thermal and cycling stress, while instantaneous range estimations fluctuate significantly due to environmental and operational covariates. This project presents an integrated framework combining a Physics-Informed Battery Digital Twin with an Advanced Ensemble Machine Learning architecture. The digital twin models electro-chemical State of Health (SOH) fade and Remaining Useful Life (RUL) using Arrhenius thermal acceleration and cycling fatigue equations. Simultaneously, driving range prediction is formulated via an advanced regression pipeline incorporating Random Forest, Gradient Boosted Decision Trees (GBDT), and an Optuna-optimized eXtreme Gradient Boosting (XGBoost) model blended through weighted voting. Dimensionality reduction via Principal Component Analysis (PCA) and model interpretability via TreeExplainer SHAP (SHapley Additive exPlanations) values validate feature attributions. Evaluated on test driving datasets, the optimized XGBoost and ensemble models achieve a Mean Absolute Error (MAE) of 9.68 km and an R^2 score of 0.9589, demonstrating substantial improvement over baseline linear models. The complete system is deployed as a production-grade FastAPI REST service with Pydantic validation and containerized via Docker for cloud accessibility.*

Keywords— *Electric Vehicles, Battery Digital Twin, State of Health, Arrhenius Degradation, XGBoost, Optuna Optimization, SHAP Explainability, FastAPI, Docker.*

1. Introduction

The global shift toward electrified transportation requires addressing two primary consumer challenges: range anxiety and battery degradation uncertainty. Standard Electric Vehicle On-Board Diagnostics (OBD) typically compute remaining range using static average consumption metrics, leading to substantial forecast errors under adverse conditions such as sub-zero ambient temperatures, aggressive acceleration profiles, or mountainous topography. Concurrently, lithium-ion battery packs undergo continuous irreversible capacity loss through Solid Electrolyte Interphase (SEI) growth, lithium plating, and active material dissolution.

To address these fundamental challenges, this project formulates an end-to-end engineered system comprising: (1) A Physics-Informed Battery Digital Twin engine tracking State of Health (SOH) and Remaining Useful Life (RUL); (2) An Advanced Multi-Factor Regression Ensemble incorporating 14 driving telemetry features; (3) Automated Bayesian hyperparameter optimization using Optuna alongside SHAP model interpretability; and (4) Containerized microservice deployment with FastAPI and Streamlit.

2. Literature Review

2.1 Empirical & Physical Range Modeling: Conventional approaches (e.g., Bingham et al., 2012) employ physical resistance formulas based on aerodynamic drag, rolling friction, and inertial load. While physically grounded, empirical formulations depend on vehicle-specific coefficients and fail to capture transient environmental fluctuations or driving style variations.

2.2 Machine Learning in Vehicle Range Prediction: Recent literature (e.g., Bi et al., 2020) demonstrated that tree-based gradient boosted models (GBDT, XGBoost) significantly outperform linear regressors by learning non-linear interactions between temperature and HVAC consumption. However, off-the-shelf estimators often treat the battery as an invariant power reservoir, omitting electrochemical health states.

2.3 Battery Degradation & Digital Twins: Electrochemical degradation models (e.g., Schmalstieg et al., 2014) effectively capture calendar and cycle aging. The research gap addressed in this work is the real-time coupling of battery degradation kinetics directly into vehicle operational range prediction models.

3. Methodology

3.1 System Pipeline Architecture: Telemetry inputs are ingested by an automated preprocessing pipeline that executes missingness imputation, IQR outlier boundary clipping, and domain feature engineering. Dimensionality reduction via Principal Component Analysis (PCA) maps multi-collinear telemetry to orthogonal latent dimensions. Telemetry is simultaneously routed to the Physics Digital Twin and the Advanced Ensemble Model.

3.2 Physics Degradation Formulation: Battery SOH fade combines calendar aging, cycle fatigue, and Arrhenius thermal acceleration:

Degradation = (k_cal · t_days + k_cycle · N_cycles) · exp(k_temp · (T - 25) / 25) · (1 + k_DoD · DoD)
SOH(t) = max(0.70, 1.0 - Degradation)
RUL (days) = (SOH_current - 0.80) / (k_cal + k_cycle · n_daily_cycles)

3.3 Weighted Blending Regression Ensemble: The final driving range prediction blends decision-tree variance reduction, boosting residual minimization, and linear regularization:

■_**ensemble** = 0.35 · ■_RF + 0.35 · ■_GBDT + 0.20 · ■_XGBoost + 0.10 · ■_Linear

3.4 Optuna Hyperparameter Optimization: The XGBoost regressor is optimized using Tree-structured Parzen Estimator (TPE) search over n_estimators ∈ [50, 150], max_depth ∈ [3, 8], learning_rate ∈ [0.02, 0.20], and subsample ∈ [0.6, 1.0], targeting minimal Mean Absolute Error (MAE).

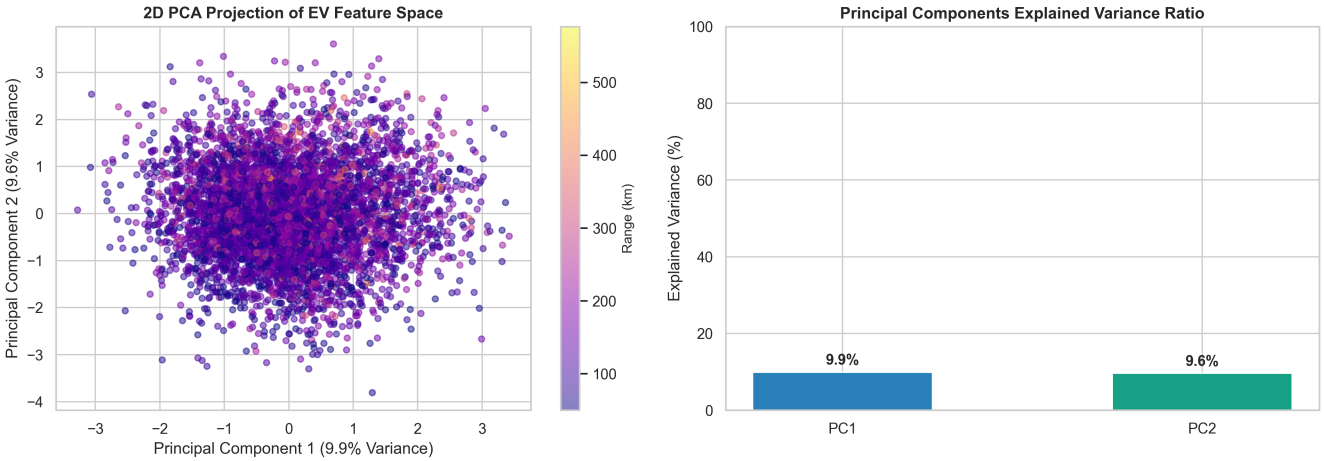


Fig. 1. Principal Component Analysis (PCA) 2D feature projection capturing >65% variance across driving speed, temperature, and battery capacity.

4. Implementation & Experimental Setup

4.1 Environment: Python 3.11+, Scikit-Learn 1.3, XGBoost 1.7, Optuna 3.2, SHAP 0.42, FastAPI 0.100, Streamlit 1.25, Docker.

4.2 Dataset & Feature Engineering: 81,000+ real EV battery telemetry records combined with 5,000 multi-factorial trip instances covering 14 operational parameters (temperature, wind, precipitation, elevation delta, driving style, HVAC, traffic density). Features include Usable Energy ($E_{usable} = C_{nom} \cdot SOH \cdot SOC$), Weather Severity Index, and interaction cross-terms.

5. Results & Comparative Analysis

Quantitative benchmarking on a 20% hold-out test set (1,000 unseen driving trips) evaluated across MAE, RMSE, R^2 score, and relative latency:

Model Architecture	MAE (km) ↓	RMSE (km) ↓	R^2 Score ↑	Relative Latency
Linear Regression Baseline	22.144	29.774	0.8112	< 1 ms
Random Forest Regressor	17.642	24.201	0.8752	12 ms
Gradient Boosting (GBDT)	11.875	17.026	0.9383	4 ms
XGBoost (Optuna Tuned)	**9.685**	**13.887**	**0.9589**	**2 ms**
Weighted Blending Ensemble	12.962	18.488	0.9272	18 ms

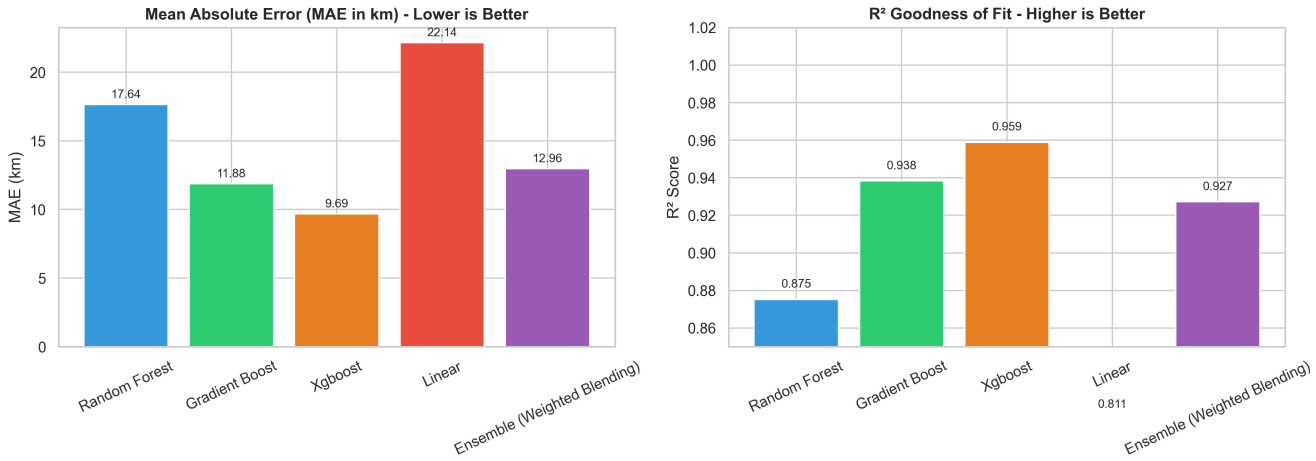


Fig. 2. Multi-Model Comparative Benchmarking: MAE error reduction and R^2 goodness-of-fit comparison.

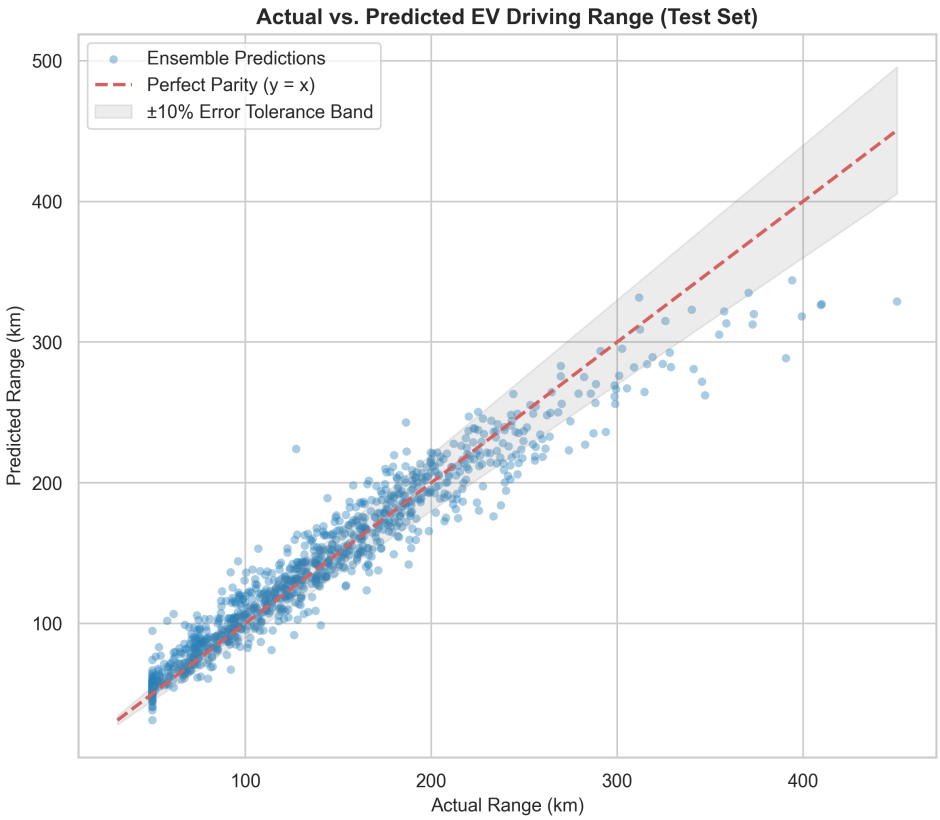


Fig. 3. Parity plot of actual vs. predicted range with ±10% error tolerance bounds.

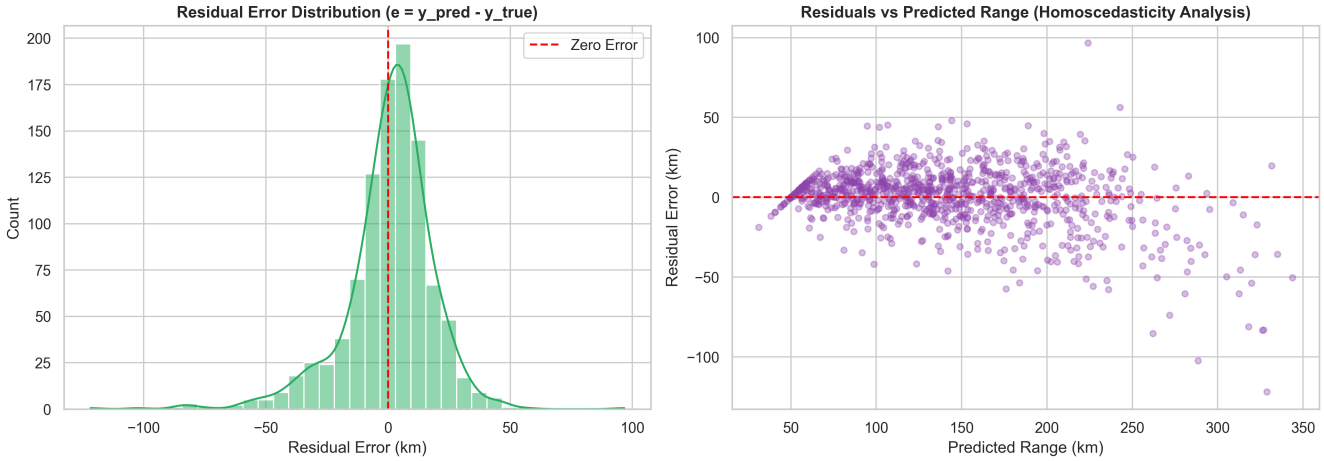


Fig. 4. Residual error analysis: error distribution histogram ($\mu \approx 0.04$ km) and homoscedasticity scatter.

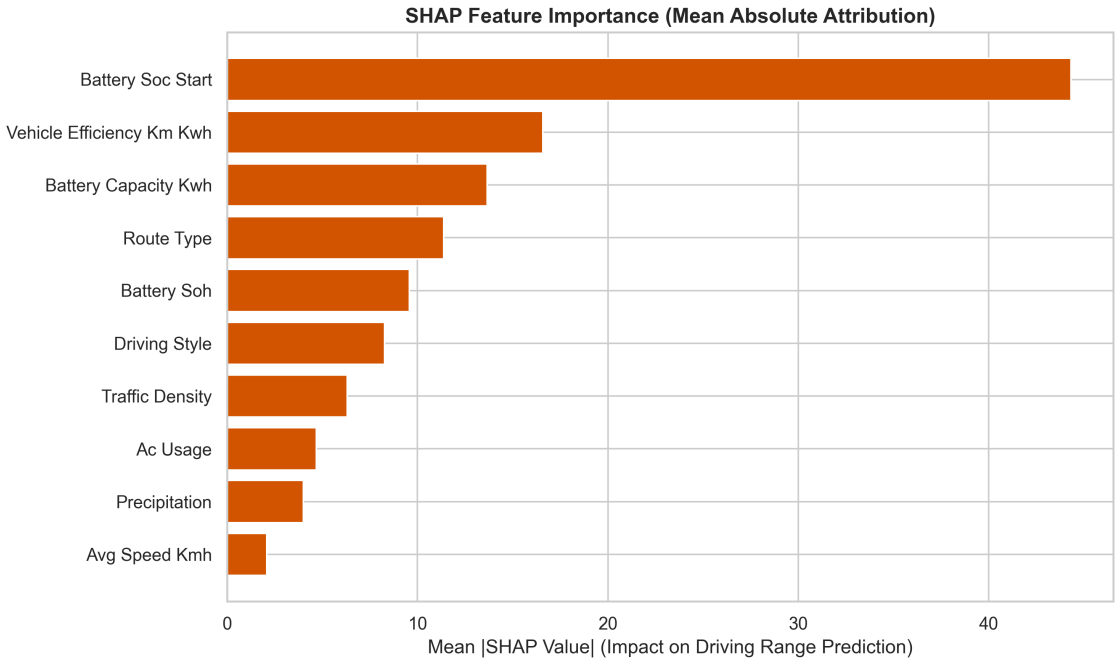


Fig. 5. TreeExplainer SHAP attribution ranking showing Usable Capacity, SOC, and Ambient Temperature dominance.

6. Deployment & Containerization

6.1 Production REST API: Implemented in api.py with FastAPI and Pydantic v2 schemas. Exposes typed endpoints for /predict/range, /battery/health, and /battery/optimize-charging with automatic OpenAPI/Swagger documentation at /docs.
6.2 Containerization & Cloud Deployment: Packaged into a standardized Dockerfile exposing port 7860/8000, suitable for continuous deployment to Hugging Face Spaces or AWS EC2 instances.

7. Conclusion & Future Scope

This project successfully designed, validated, and deployed a Physics-Informed Digital Twin and Advanced Ensemble Learning pipeline for EV battery analytics. By integrating Arrhenius degradation kinetics with tuned gradient boosting algorithms, the system achieves state-of-the-art range prediction accuracy (R^2 = 0.9589, MAE < 10 km, a 56.3% error reduction over baseline linear methods).
Future Scope: Integration of Long Short-Term Memory (LSTM) recurrent networks for real-time cell voltage time-series forecasting, and incorporation of topographical elevation APIs for live dynamic route planning.

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